

1 Data Preparation

- TriviaQA rc.nocontext
- 2,000 train / 200 eval questions
- K_0 factual baseline (held-out)
- Bidirectional substring matching

2 Recursive Training

- QLoRA 4-bit NF4 / FFT
- Fresh adapter each generation
- 10 generations $G_1 \rightarrow G_{10}$
- 3 backbones: Qwen, Gemma 3, 4

$\hookrightarrow$ Output Gen $t \rightarrow$ training for Gen $t+1$

3 Pressure Axes

- Adapter rank — 4, 16, 32, 64, 128, 256
- Learning rate — $1e-6$ to $2e-5$
- Rank $\times$ LR joint-dependence grid
- Exposure — 0/10/25/50% removal

4 Multi-Metric Evaluation

- K_0 retention — factual preservation
- Output drift — Distinct-n, MTLD, stopword ratio
- Effective rank — $\exp(H(\sigma))$ of $B \times A$

5 Regime Classification

Homeostatic — retention > 90%, stable

Bounded
— stable retention, distribution drifts

Degradative
— retention declines ~ 0.9 – 2.5 pp/gen

6 Pressure-Regime Mapping

- Effective training pressure (ETP) — boundary identification
- Regime transitions: threshold-like within grid
- Cross-backbone generalization
- Rank, LR, exposure $\rightarrow$ unified pressure